

Unit 3 - Web Analytics

Digital Marketing (DI03016011) - GTU Diploma IT Sem 3

22% weightage | 6 hours

How to use these notes: each topic is written the way you can copy it in the exam. **3 marks** = definition + 2-3 points. **4 marks** = add example or table. **7 marks** = write everything, with diagram/table and a real example. Lines marked ★ show where the topic was asked in a GTU paper (only 2 papers exist: Winter 2025 and Summer 2026).

Questions already asked from this unit (do these first)

- Unique visitors asked in W2025 Q3(a) [3M] + S2026 Q3(a)OR [3M]
- W2025 Q3(a)OR [3M: Pages/Visit and New Visits]
- S2026 Q3(a)OR [3M: unique visitors, average visit duration, bounce rate]
- BOTH papers: W2025 Q3(b) [4M: weblogs vs page tagging] + S2026 Q4(b)OR [4M: methods of data collection]
- S2026 Q4(a) [3M: define analytics + list SEO analytics tools]
- BOTH papers (7M each): W2025 Q3(c) [A/B testing and difference from multivariate] + S2026 Q3(c) [A/B and multivariate testing tools and their role]
- W2025 Q4(a) [3M: conversion rate and its importance]
- S2026 Q3(c)OR [7M: how to set up goals in Google Analytics, with example]
- S2026 Q3(a) [3M: any one single-touch attribution model with example]
- W2025 Q3(c)OR [7M: compare Linear and Time-Decay multi-touch attribution models with examples]
- BOTH papers: W2025 Q3(b)OR [4M: what is GTM, how it simplifies setup] + S2026 Q4(a)OR [3M: define GTM and list tasks]

3.1 Key metrics: unique visitors

★ **Unique visitors asked in W2025 Q3(a) [3M] + S2026 Q3(a)OR [3M]**

- **Unique visitors:** the number of **different individual people (or devices)** who visit a website in a given period. Each person is counted **only once**, even if he visits 10 times (identified by cookie / device ID).
- **Why it matters:** shows the real size of the audience and reach of the site, unlike total visits.

Example: Ravi visits your site 5 times and Meena 2 times in a week → **visits = 7, unique visitors = 2.**

Pageviews (related): total number of pages viewed; one page opened again is counted again.

3.1 Key metrics: pages/visit, new visits

★ **W2025 Q3(a)OR [3M: Pages/Visit and New Visits]**

- **Pages/Visit** (pages per session): the **average number of pages** a visitor views in one visit. Higher value = visitors are interested and exploring.

Formula: Pages/Visit = Total pageviews ÷ Total visits. Example: 5,000 pageviews ÷ 2,000 visits = **2.5**.

- **New Visits** (% new visits): the **percentage of visits made by first-time visitors**. Shows how well the site attracts new audience; the rest are returning visitors (loyalty).

Formula: % New visits = (Visits from new visitors ÷ Total visits) × 100. Example: 600 of 1,000 visits are new → **60%**.

3.1 Key metrics: average visit duration, bounce rate

★ **S2026 Q3(a)OR [3M: unique visitors, average visit duration, bounce rate]**

- **Average Visit Duration:** the average time a visitor spends on the site in one visit. Longer time usually means interesting content.

Formula: Avg visit duration = Total time spent by all visits ÷ Number of visits. Example: 6,000 sec ÷ 100 visits = 60 sec.

- **Bounce rate:** the percentage of visits in which the visitor **leaves after viewing only one page** without any interaction. High bounce rate = page is not relevant, slow or confusing.

Formula: Bounce rate = (Single-page visits ÷ Total visits) × 100. Example: 40 of 100 visits leave from the first page → **40%**.

Note: in GA4, bounce rate is the opposite of 'engagement rate' (sessions that were not engaged).

3.1 Data collection: weblogs and page tagging

★ **BOTH papers: W2025 Q3(b) [4M: weblogs vs page tagging] + S2026 Q4(b)OR [4M: methods of data collection]**

Two main methods are used to collect website visitor data:

Point	Weblogs (server log files)	Page tagging (JavaScript tags)
How it works	Web server automatically records every request in a log file (IP, date/time, page, status code, browser, referrer)	A small JavaScript code (tag) on each page runs in the visitor's browser and sends data to the analytics server
Data location	On your own server	On the analytics vendor's server (e.g. Google Analytics)
Setup	No code needed on pages; needs log analyser software	Add tag to every page (or use Google Tag Manager)
Tracks	Page requests, bots and crawlers, errors (404)	User behaviour: clicks, scrolls, events, time on page, device, real-time
Limitations	Cannot track JavaScript events; cached pages and bots confuse counts; hard to identify unique visitors	Fails if JavaScript is disabled or blocked (ad-blockers); depends on third party
Example	Apache / IIS log analysis	Google Analytics, Matomo

Other methods: cookies (identify returning visitors), packet sniffing, and in-app SDKs for mobile apps.

3.2 Web analytics tools and SEO analytics tools

★ **S2026 Q4(a) [3M: define analytics + list SEO analytics tools]**

Web analytics: the collection, measurement, analysis and reporting of website data (visitors, pages, sources, conversions) to understand user behaviour and improve the website and marketing.

SEO analytics tools

- **Google Search Console** - clicks, impressions, ranking queries, indexing errors.
- **Google Analytics** - organic traffic and behaviour.

- **Ahrefs, SEMrush, Moz** - keywords, backlinks, rank tracking.
- **Ubersuggest, SEOptimer, Sitechecker** - audits and keyword ideas.
- Google PageSpeed Insights / GTmetrix - speed.

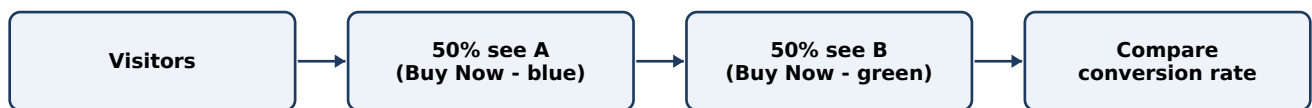
Types of web analytics tools (syllabus 3.2)

Type	Purpose / example
Content analytics	Which pages / content perform best
Customer analytics	Who the customers are and how they behave
Usability (UX) analytics	Heatmaps, session recordings - Hotjar
A/B and multivariate testing	Compare versions - Optimizely, VWO
SEO analytics	Search Console, Ahrefs, SEMrush
General enterprise analytics	Google Analytics, Adobe Analytics
Open-source analytics	Matomo, Open Web Analytics
Product analytics	In-app behaviour - Mixpanel, Amplitude

3.2 A/B and multivariate testing

★ **BOTH papers (7M each): W2025 Q3(c) [A/B testing and difference from multivariate] + S2026 Q3(c) [A/B and multivariate testing tools and their role]**

- **A/B testing (split testing):** two versions of a page / element - **A (control)** and **B (variation)** - are shown to **different random groups** of visitors at the same time. The version with the better result (more clicks, sign-ups, sales) wins. Only **one element** is changed (e.g. button colour or headline).
- **Multivariate testing (MVT):** tests **several elements together** and all their **combinations** to find the best combination.



Point	A/B testing	Multivariate testing
What is tested	One variable, two (or few) versions	Multiple variables and their combinations
Example	Headline 1 vs Headline 2	2 headlines × 2 images × 2 buttons = 8 combinations
Traffic needed	Less	Much more (traffic is divided among all combinations)
Time / complexity	Simple and fast	Complex, longer
Result	Which version is better	Which combination, and how elements interact
Best for	Big changes, low-traffic sites	Fine-tuning high-traffic pages

Testing tools and their role in optimising website performance

- **Tools:** Optimizely, VWO (Visual Website Optimizer), AB Tasty, Adobe Target, Convert (Google Optimize was shut down in 2023).
- **Role:** split traffic randomly, show variations, track goals, calculate **statistical significance** and show the winner.
- **Benefits:** increases **conversion rate**, decisions based on data not guesses, lowers risk of bad changes, improves user experience and ROI.

3.3 Conversion rate

★ **W2025 Q4(a) [3M: conversion rate and its importance]**

Conversion rate is the percentage of visitors who complete a **desired action (goal)** such as a purchase, sign-up, form submission or download.

Formula: Conversion rate = (Number of conversions ÷ Number of visitors) × 100. Example: 50 sales from 2,000 visitors = **2.5%**.

Why it is important

- Shows how **effective** the website / campaign is in turning visitors into customers.
- Helps find weak pages and improve them (A/B testing).
- Lets you compare channels and calculate **ROI**; improves profit without extra traffic cost.

3.3 Setting up goals in Google Analytics

★ S2026 Q3(c)OR [7M: how to set up goals in Google Analytics, with example]

A **goal** measures how often visitors complete a valuable action. Setting goals lets you calculate the **conversion rate**. (In GA4 the older 'goals' are set up as **events marked as key events / conversions**; menu names may differ slightly by version.)

Common goal types

Goal type	Meaning	Example
Destination	Visitor reaches a specific page	'Thank you' page after order (/thank-you)
Duration	Session lasts longer than a set time	Visit longer than 3 minutes
Pages / screens per session	Visitor views at least N pages	Views 5 or more pages
Event	A specific action happens	Click on 'Add to cart', video play, form submit

Steps (example: online cake shop wants to measure orders)

1. Connect the website to **Google Analytics** (tag directly or through GTM).
2. Open **Admin** → Events (or Goals) and choose **create new goal / event**.
3. Select the type - e.g. **Destination** goal with page */order-confirmation* (or an event 'purchase').
4. Give it a name ('Cake order'), optionally a **value** (e.g. ₹800) and a funnel (cart → checkout → confirmation).
5. Save / mark the event as a key event (conversion) and **test** it.
6. View **goal / conversion reports**: number of conversions, **conversion rate**, source that brought them.

Result: If 40 of 1,000 visitors reach the confirmation page, conversion rate = 4%; the report also shows whether Instagram or Google gives more orders.

3.4 Attribution: single-touch model

★ S2026 Q3(a) [3M: any one single-touch attribution model with example]

Attribution model: a rule that decides **which marketing touchpoint gets credit** for a conversion. **Single-touch** models give 100% credit to just one touchpoint.

- **First-touch (first-click):** all credit goes to the **first** interaction that introduced the customer.
- **Last-touch (last-click):** all credit goes to the **last** interaction before the conversion.

Example (last-click): A customer sees an Instagram ad → reads an email → searches the brand on Google and clicks a search ad → buys a ₹1,000 product. Under last-click, the **Google search ad gets full ₹1,000 credit**; Instagram and email get 0. Under first-touch the Instagram ad would get it.

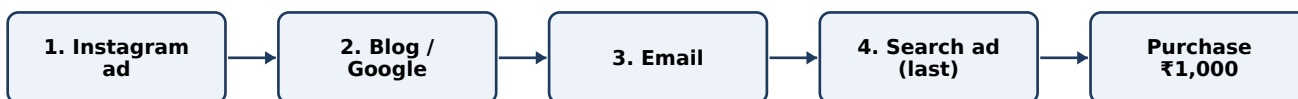
Weakness: ignores all other touchpoints.

3.4 Multi-touch attribution: linear vs time decay

★ W2025 Q3(c)OR [7M: compare Linear and Time-Decay multi-touch attribution models with examples]

Multi-touch attribution shares the credit for a conversion among **several touchpoints** in the customer journey.

Customer journey example: sale of ₹1,000 with 4 touchpoints in order: ① **Instagram ad** → ② **Blog post (Google search)** → ③ **Email** → ④ **Search ad click** → **Purchase**



Linear model

Gives **equal credit** to every touchpoint. Each gets 25% = **₹250**. Good for long journeys where every step helps; ignores that some touches matter more.

Time-decay model

Gives **more credit to touchpoints closer to the conversion**, less to older ones. Example split: 10% / 20% / 30% / 40% = **₹100, ₹200, ₹300, ₹400**. Good for short sales cycles or promotions.

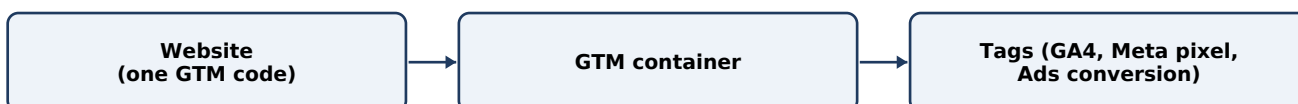
Point	Linear	Time-decay
Credit	Equal to all touchpoints	More to recent touchpoints
Example (₹1,000)	₹250 each	₹100 / ₹200 / ₹300 / ₹400
Best for	Long journeys; brand building	Short campaigns, flash sales
Drawback	Treats all touches as same	Undervalues the first (awareness) touch

Other multi-touch models: position-based (U-shaped), data-driven. Note: newer GA4 versions offer mainly data-driven and last-click models; the syllabus models are still asked in theory.

3.5 Google Tag Manager

★ **BOTH papers: W2025 Q3(b)OR [4M: what is GTM, how it simplifies setup] + S2026 Q4(a)OR [3M: define GTM and list tasks]**

Google Tag Manager (GTM) is a **free tag-management system** by Google. You place **one GTM container code** on the website, and then add, edit and manage all **tags** (small tracking code snippets such as Google Analytics, Meta pixel, conversion tracking) from a web dashboard **without changing the website code** each time.



Main parts

- **Tag:** code that sends data to a tool.
- **Trigger:** the rule when the tag fires (page view, button click, form submit).
- **Variable:** values used by tags / triggers (page URL, click text).

Tasks of GTM / how it simplifies analytics setup

- Add and update tracking tags **without a developer**.
- Track clicks, form submissions, scroll, video views, downloads (event tracking).
- Manage many tags in **one place** - faster page loading than many separate scripts.
- **Preview / debug mode** to test before publishing, with **version control** and rollback.
- Set up conversion tracking for ads and remarketing.

Last-minute revision sheet - Unit 3

- Unique visitors = people counted once. Pages/visit = pageviews ÷ visits. Bounce = single-page visits ÷ visits.
- Weblogs = server records; page tagging = JavaScript tag.
- Conversion rate = conversions ÷ visitors × 100.
- A/B = 1 change, 2 versions; MVT = many elements and combinations (more traffic).
- Linear = equal credit; time decay = more credit to recent touches. First/last-touch = single touch.
- GTM = manage tags without editing code: tag, trigger, variable.